EVM ANALYSIS: SPI, CPI, EAC & ETC EXPLAINED

Earned Value Management in Construction Projects

1 Schedule Performance Index

SPI = EV ÷ PV

- ✓ Ahead of Schedule (SPI > 1.0)
- ✓ On Schedule (SPI = 1.0)
- ✓ Behind Schedule (SPI < 1.0)

2 Cost Performance Index

CPI = EV ÷ AC

- ✓ Under Budget (CPI > 1.0)
- ✓ On Budget (CPI = 1.0)
- ✓ Over Budget (CPI < 1.0)

3 Estimate at Completion

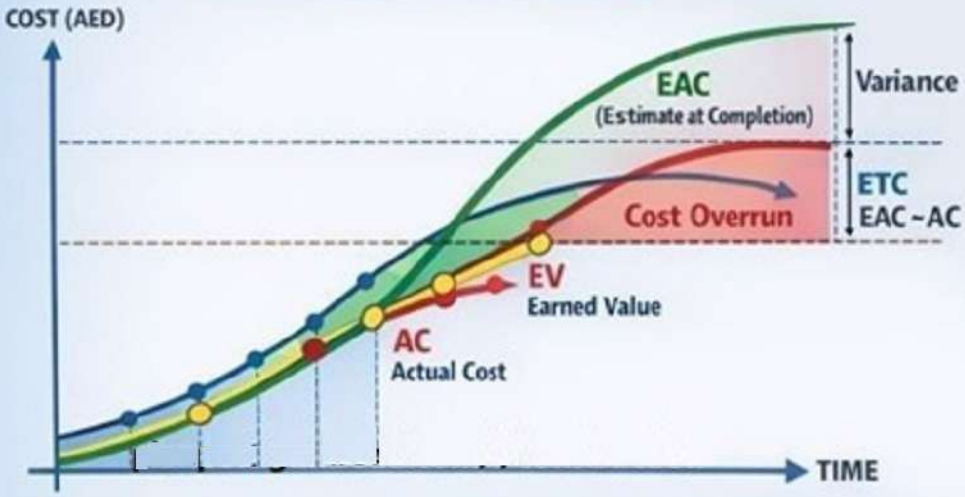
Forecasted Total Cost

4 Estimate to Complete

Cost to Finish Remaining Work



EVM FORECASTING GRAPH



- PV Planned Value
- EV Earned Value
- AC Actual Cost
- EAC Forecasted Cost

- ▶ EV < PV = Schedule Delay (SPI < 1)
- ▶ AC > EV = Cost Overrun (CPI < 1)